

# Learning objectives

In this lesson you will learn to:

- Define what is meant by the term 'cyberattack'
- Describe the financial, reputational and legal damage that a cyberattack can cause
- Describe the characteristics of and threats posed by different types of malware
- Describe how anti-malware software works
- Explain why it is important to keep anti-malware software up to date.

For more information and additional learner activities see Chapter 5.3 of the student book.

## Twitter under attack

In mid-July 2020, a number of high-profile verified Twitter accounts were breached. Accounts associated with Barak Obama, Jeff Bezos, Bill Gates, Apple and Uber all posted similar tweets instructing people to send bitcoin to the same account. The tweets were removed, but they continued to be posted throughout the day.

For a day, verified accounts were unable to post tweets as Twitter investigated the attack.

# Cyberattack

The news story we have just read is an example of a large scale **cyberattack**.

Cyberattacks are malicious acts in which a computer system comes under attack by unauthorised persons, who are generally known as hackers.

Why might people launch such attacks?

## Financial and reputational damage

WannaCry was a **ransomware** attack that affected hundreds of organisations in 2017. The National Health Service (NHS) suffered an estimated £92-million loss as a direct result of the attack. The NHS also suffered damage to its reputation.

WannaCry is a known vulnerability that required a patch to fix. Microsoft released the patch, but many NHS trusts did not implement it. This meant that they were vulnerable to the attack. When this came to light, many people were concerned about the lack of up-to-date software within the NHS.

## Legal damage

Attacks of this nature can have legal implications for the organisations that are impacted by them.

In May 2020, the airline easyJet came under attack, when the email addresses and travel details of approximately 9 million customers were stolen. More than 2,000 customers also had their credit card details stolen.

In June 2020, over 10,000 people joined a class-action lawsuit against easyJet in an attempt to get compensation for the loss of their data.

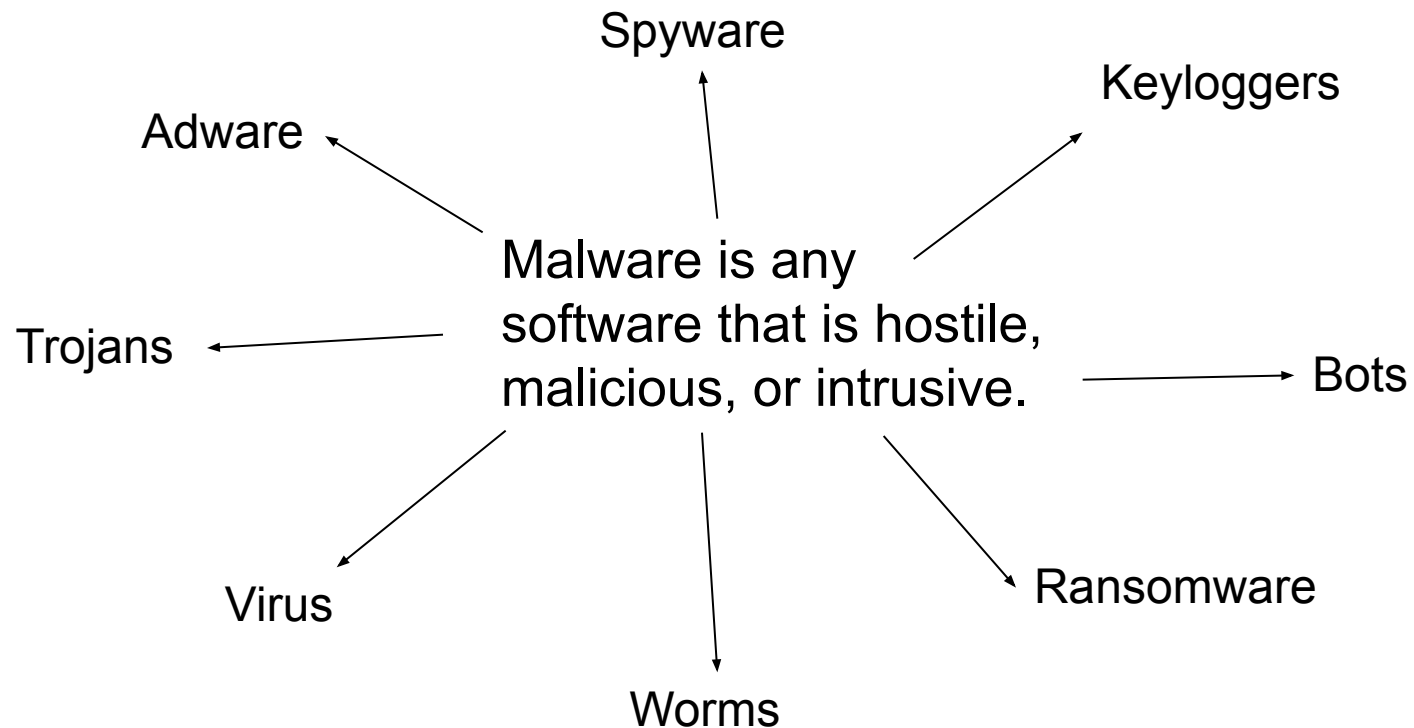
## Targeting information

Often the purpose of cyberattacks is to steal information. This is usually data such as email addresses and credit card details. This information is often used for further crime such as identity theft or fraud.

Information that is stolen might also comprise intellectual property, such as source codes or patented designs that can be sold on the black market.

Sometimes data is encrypted (using ransomware attacks) in order to damage an organisation financially or reputationally.

# What is malware?



# Anti-malware

One of the best defences against malware is **anti-malware** software.

This is software designed to detect, quarantine, and destroy potentially damaging programs.

When a file enters a computer system, anti-malware software will scan the file for known malicious code.

If the anti-malware software finds a problem, the malicious program is quarantined so that it cannot affect the system. The user is then notified and can decide to let in the problematic software or delete it.

## Known code? Heuristic analysis?

Anti-malware programs can only work with information that is already known about malicious programs.

However, new malware is being created all the time to exploit different vulnerabilities. It is becoming more sophisticated and less easy to identify.

As a result, it is vital that anti-malware protection is kept up to date, so that known malware can be identified reliably.

But what if a piece of malware is brand new?

# Wrap up: you have learned how to...

- ✓ Define what is meant by the term 'cyberattack'.
  - Digital attack on a computer system.
- ✓ Describe the financial, reputational and legal damage that a cyberattack can cause.
  - The cost of lost business and reputational damage, the impact of legal cases and the cost of implementing software fixes.
- ✓ Describe the characteristics of and threat posed by different types of malware.
  - Viruses, worms, trojans, spyware, and ransomware.
- ✓ Describe how anti-malware works.
  - It detects, quarantines and destroys malicious code.
- ✓ Explain why it is important to keep anti-malware up to date.
  - New malware is being created all the time, so anti-malware software must be kept up to date so it can recognise the malicious code.